Twenty sectors, five sensor bets worth testing

The best openings for a solo, privacy-first hardware founder are not in any one sector. They are in a single repeated pattern: **a cheap, battery-powered radio sensor (ESP32/Nordic plus LoRa or BLE) that reports to a local hub, sold for a one-time price into a category where incumbents charge per device per month.** That pattern scores 19-21 out of 25 in at least eight sectors. It covers stock-tank levels, self-storage unit doors, short-term-rental noise, walk-in coolers, leak/freeze pucks, machine run-time clamps, proof-of-visit tags and school aerosol sensors. Of the roughly 100 ideas screened, only five have real-user evidence behind them. In order of evidence strength they are **proof-of-visit tags for lawn and cleaning crews** (LawnSite operators repeatedly fight "you didn't mow" non-payment), **school vape/aerosol sensors** (news reports of false-alarm floods, with a median **$925 per detector** in district contracts), **self-storage door sensors** (operators already buy unit alarms but complain about batteries and false alarms), **no-subscription STR noise monitors** (Trustpilot and host-forum complaints about Minut billing) and **clamp-on CNC monitors** (incumbents charge **$240-$1,200 per machine per year**, vendor figures). The top-scoring idea on paper, a **solar LoRa stock-tank monitor (21/25)**, rests only on a price gap: rivals charge $400-$550 plus $99-$420 a year, but no rancher voice was found. A pass through the forums listed at the end is the cheapest next step. Automated research could not open Reddit, Airbnb Community, Practical Machinist, Self-Storage Talk or LawnSite, so most demand claims remain untested.

Eighteen ideas clear 19 points, but five carry real-user evidence

Ranking every idea by adjusted score gives a crowded top. **Eighteen ideas score 19 or more**, and they cluster into three product families: radio sensor nodes with a local hub, sound-level devices, and design-led display objects. The scores mostly reflect how buildable and on-brand an idea is, since buildability and fit are the easiest criteria to score high without market evidence. The evidence column is the real filter.

Rank	Idea (sector)	Score	Evidence status
1	Stocktank LoRa (11)	21	**Unproven demand.** Price gap documented: incumbents charge $400-550 + $99-420/yr

Rank	Idea (sector)	Score	Evidence status
			(Gallagher, Tank Toad). No rancher voice found. Assumes a house within LoRa range.
2	Unit Door Sentinel (48-49)	20	**Real-user evidence.** Self-Storage Talk operators run unit alarms that catch intruders, and complain about battery upkeep and false alarms (SST, snippet only). WTP for retrofit at $50+/door unproven.
3	STR noise + occupancy puck (72)	20 [adj]	**Real-user evidence** on price model: Trustpilot billing complaints and a 6-property host saying fees add up. Competition entrenched (Minut + Airbnb).
4	Clamp-on machine monitor (31-33)	20	**Partial.** Incumbent pricing is high (vendor-authored). A Practical Machinist snippet reports $25K controller unlock fees and warnings against putting old machines online, which supports local-first (PM). Guidewheel already does clamp-on.
5	Fence Watch (11)	20	Not validated
6	Proof-of-protection log (52)	20	Not validated; insurer acceptance unknown
7	Climbing comp timer (71); bar/cafe sound sign (72); STR quiet lamp (72)	20	Not validated; sign demand is the researchers' idea with no ordinance evidence
8	Crew Beacon Tag (56)	19 [adj components]	**Strongest real-user evidence.** Repeated LawnSite threads on non-payment and crew verification (1, 2). Must beat free phone photos.
9	Quiet-Room aerosol sensor (61)	19	**Strong evidence** from news and procurement: false-alarm floods, $925 median unit, one-time settlement funding. No evidence that no-cloud matters to buyers.
10	Walk-in Watch (44-45)	19 [adj]	**Thin.** Vendor loss figures only; local alarms and no-subscription rivals exist.
11	Room puck (54)	19	**Partial.** Teams panel check-in bug thread; platform fix "coming soon".
12	Trail/park counter (92); daycare CO2 display (62); hangboard logger (71); venue SPL sign (71); laundry cycle sensor (81)	19	Not validated. The trail counter has the clearest incumbent price gap ($4-6K units).

Read with the evidence, the shortlist is different from the score order. **Crew Beacon Tag and Unit Door Sentinel** have operators describing the problem in their own words and an existing habit of paying for a fix. The **STR puck** has customers angry about exactly the subscription model the founder would avoid, which is the clearest product-market wedge in the screen.

However, it fights an incumbent that Airbnb subsidizes. The **school aerosol sensor** has the most money and the most documented dissatisfaction. It is also the hardest to win on quality, and its procurement cycles are slow. The **Stocktank** and **machine monitor** ideas score high on fit and build, but they need a few hours of forum reading before any design work. Several ideas share one hardware platform: a LoRa node, a local hub and an e-paper display. That platform covers stock tanks, unit doors, walk-ins, leak pucks and rural water tanks. Building it once lets the founder test several markets cheaply.
Three earlier favourites lost ground in validation. **Leak and freeze sensors for landlords and insurers** fell from 20 to 17-18 because hardware sells for $13-$55, insurers give devices away, and no landlord voice was found. **Closet Sentinel** fell to 18 because the $145-$500 band is crowded. **Walk-in Watch** fell to 19 because local, no-subscription alarms already exist.

Conclusion

The screen shows a design pattern more than it ranks sectors. In sector after sector, the incumbent charges a monthly fee for a radio sensor and a cloud dashboard, and the documented complaints are about that fee, about devices that stop working when the plan lapses, or about false alarms. A founder who sells reliable, local-first hardware for a one-time price is attacking the business model, not the sensor. That is why the same LoRa-node-plus-hub architecture keeps scoring 19-21 across agriculture, storage, hospitality, retail and property. The trap is that the scoring rewards what this founder is good at (buildable, on-brand) more than what the market has proven. Of eighteen ideas at 19 or above, only five have any user voice behind them.

The practical implication is to sequence by evidence, not by score. Spend a weekend in the LawnSite, Self-Storage Talk, Practical Machinist and Airbnb threads above before choosing. Favour the idea where operators already pay for a worse solution: storage door alarms, guard and crew verification, and STR monitors. Prototype the shared LoRa node and hub once, so the stock-tank and walk-in variants become cheap follow-on tests rather than separate bets. Ideas resting on vendor-only pain (walk-ins, closets, ghost bookings) or on insurer channels (leak pucks) should wait until a real buyer says, in their own words, that they would pay.

Forums the founder should read personally

Automated research hit hard limits. Reddit returned a domain error, and site-restricted searches were ignored. Airbnb Community and most of BiggerPockets were not indexed. **Practical Machinist and Self-Storage Talk returned 403 errors, and LawnSite redirected to a paywall (402).** Only search snippets of those threads were seen, and their dates are

unverified. The table below lists the specific threads the notes identified, plus the communities the researchers searched but could not reach. Reading them with a normal browser, and noting dates, quotes and prices paid, is what would turn the "unproven" labels above into evidence.

Community	What to read	Tests which idea
LawnSite	"Customer claims I didn't cut his grass"; "Customer claims their lawn hasn't been mowed"; "How do you verify that employee cut the customer's lawn?"; "How to respond to BS customer complaints"	Crew Beacon Tag: dispute frequency, dollar loss, whether photos already suffice
Self-Storage Talk	"Individual Unit Alarms"; "Individual door alarms"; fence security thread	Unit Door Sentinel: per-door prices paid, battery and false-alarm complaints
Practical Machinist	"Is connecting machines to the network worth it"; "Shop floor data collection"; "MTConnect: two shops share their experience"	Clamp-on machine monitor: subscription resistance, local-first demand
Airbnb Community	Host threads on noise monitors, Minut/NoiseAware billing, party damage (no specific threads retrievable; search community.withairbnb.com)	STR puck: willingness to pay a higher one-time price
Reddit	r/airbnb_hosts (STR puck); r/landlord, r/PropertyManagement (leak sensors); r/Teachers, r/k12sysadmin (vape sensor); r/restaurateur, r/restaurantowners, r/KitchenConfidential (walk-in alarms); r/ranching, r/homestead (stock tank); r/Machinists (machine monitor); r/selfstorage (door sensor); r/sysadmin, r/msp, r/homelab (closet monitor); r/facilitiesmanagement, r/Office365 (room puck); r/CommercialCleaning, r/securityguards, r/sweatystartup (proof-of-visit); r/Truckers (trailer tracker); r/gymowner (gym access); r/AutoRepair, r/laundromat (sector 81); r/ECEProfessionals (daycare); r/podcasting (OnAir puck); r/LocalGovernment (counters); r/Construction (jobsite)	Every idea; Reddit was fully blocked
Other forums worth a look	STR Specialist host thread (readable); eMastercam monitoring thread; Level1Techs server-room thread; Edugeek server-room thread; Microsoft Teams panels thread; OBS on-air light thread; Spiceworks and ranchers.net (searched, nothing current)	Closet Sentinel, room puck, OnAir puck, Stocktank

All 20 sectors

Number on the right is the sector's top idea score.

11	Agriculture, Forestry, Fishing and Hunting	21	21	Mining, Quarrying, and Oil and Gas Extraction	16	22	Utilities	18
23	Construction	18	31-33	Manufacturing	20	42	Wholesale Trade	18
44-45	Retail Trade	17	48-49	Transportation and Warehousing	20	51	Information	18
52	Finance and Insurance	20	53	Real Estate and Rental and Leasing	16	54	Professional, Scientific, and Technical Services	19
55	Management of Companies and Enterprises	17	56	Administrative and Support and Waste Management	16	61	Educational Services	19
62	Health Care and Social Assistance	19	71	Arts, Entertainment, and Recreation	20	72	Accommodation and Food Services	20
81	Other Services	19	92	Public Administration	19			

NAICS 11

Agriculture, Forestry, Fishing and Hunting

SUMMARY

About 1.88 million mostly small farms buy inputs through dealers, co-ops and online stores. Margins are volatile, so tools that pay back within two years sell. Remote monitoring of water, fences and grain where there is no coverage is the clearest electronics gap. Incumbents such as Gallagher are expensive or charge subscriptions.

KEY FACTS

114,570 establishments, 1,259,402 employees, $255.9B value added (2025) (QCEW 11). QCEW misses most farm operators. NASS counts **1.88M farms in 2024**, averaging 466 acres (NASS). Net farm income was forecast at $140B for 2024 (DTN). Key rules: FCC Part 15 for

unlicensed radios, which pre-certified modules cover, and fence-energizer standards (both from the researchers' reading; no source retrieved).

BUYERS

Segment	Count	How they buy
Small/hobby and mid-size livestock and crop farms	Most of 1.88M farms; split by size not found	Online and farm stores; owner decides; price-sensitive
Commercial ranches and large grain operations	Not found	Ag dealers, co-ops, extension-recommended brands; accept subscriptions if payback is clear
Ag service channel (irrigation dealers, electric co-ops)	Not found	Resell and install

PAIN POINTS

Pain	Cost	Current workaround
Checking remote stock tanks	Trips about **$1,560/yr**; monitor $1,800 + $100 install + $3-10/mo, ~2-yr payback if trips halve (USU Extension)	Drive out; cameras; cellular or satellite monitors (~$30/mo satellite, UNL)
Stored-grain spoilage	Loss per bin not found; cables ~5 cents/bushel (The Farmer)	Bin cables; some vendors charge for remote access
Electric-fence faults let stock out	Cost of escapes not found	Gallagher monitors ($277.99-$349.99)
Pump motors damaged by single-phasing	Not found	Non-connected phase-monitor relays (UC ANR)
Income volatility caps spending	Context only	Delay purchases

EXISTING PLAYERS

Company	Product	Price	Gap
Gallagher	Fence monitor; tank monitor (cellular/satellite)	Fence monitor $349.99, fault finder $277.99 (Gallagher); tank monitor $399.99 + $99/yr (unverified)	Costly; ecosystem lock-in
Ranchbot	Trough Direct (satellite)	$549 + ~$270/yr (unverified) (Feedlot)	Subscription
Tank Toad	Solar sonar tank monitor	Cellular renewal $420/yr (Solar Store)	Subscription
Lonestar, Farmbot, Farmo, Goanna	Tank monitors	Not found	Subscription models

Company	Product	Price	Gap
Guardian, TeleSense, OPI, Prairie Grain	Grain-bin monitoring	~5 c/bu; remote access sometimes paid	Subscription for remote access

PREAS

1 Stocktank LoRa

21/25
P/W/O/B/F 4/4/4/5/4

tank-level node and solar gateway on LoRa, local display, no cloud

B2B
Ranches pay to cut trips; $150-250/node + optional $5/mo alerts

B2C
Hobby farmer, ~$99 node + $79 gateway

`Easy build · ESP32/LoRa, ultrasonic or 4-20mA pressure sensor, e-paper, solar + LiPo, optional cellular gateway`

2 Fence Watch

20/25
P/W/O/B/F 3/4/4/5/4

clamp-on fence-voltage sensor radioing to a base display

B2B
Ranches; $60/node + $10 per gateway

B2C
$59

`Easy build · High-voltage divider, LoRa, e-paper`

3 Pasture Radio Ear

17/25 [stated 15]
P/W/O/B/F 2/2/4/4/5

local "farm dashboard" aggregating all nodes, no cloud

B2B
None (B2C only)

B2C
$149

`Easy-moderate build · ESP32, LoRa, display`

4 Bin Probe

16/25 [stated 18]
P/W/O/B/F 4/3/2/4/3

wireless temp/CO2 puck for bins, no subscription

B2B
Grain operations; $40-80/probe

B2C
Same, $40-80

`Moderate build · Thermistor strings, NDIR CO2, BLE/LoRa`

5 PhaseGuard Mini

15/25 [stated 18]
P/W/O/B/F 3/3/3/3/3

pump phase-loss relay with LoRa alert

B2B
Irrigation dealers, $120-200

B2C
Not specified

`Moderate (mains switching raises UL concerns) build · Phase sensing, relay, LoRa`

NAICS 21

Mining, Quarrying, and Oil and Gas Extraction

SUMMARY

Marginal oil and gas wells and small quarries are numerous, have thin margins and are driven by compliance. Remote tank and well monitoring is a proven paid category with high incumbent prices. Hazardous-location certification is the main barrier for a solo founder (researchers' inference), so the realistic entry is a non-hazardous water or fuel tank variant.

KEY FACTS

30,746 establishments, 587,151 employees, $398.7B value added (QCEW 21). There were about **770,000 marginal wells in 2016**, producing 11-15% of US oil and gas; a current count was not found (EIA). MSHA counts **12,721 mines and 328,732 miners** in FY2024 (MSHA). Key rules: EPA OOOOb/c methane rules, with most deadlines pushed to Jan 22, 2027 and reconsideration ongoing (sources conflict on dates) (Gable Law); MSHA dust rules; Class I Div 1/ATEX hazardous-location rules (no source retrieved).

BUYERS

Segment	Count	How they buy
Small independent oil/gas operators	Not found (over 50% of marginal-well output is from private operators, Argus)	Owner or pumper decides; oilfield supply houses and SCADA resellers
Quarry and aggregate operators	~4,286 stone + 6,198 sand and gravel (2023, NIOSH)	Plant manager or safety lead; distributors
E&P compliance teams and leak-detection contractors	Not found	Per site; contractor LDAR ~$250/site (advocacy source, EDF)

PAIN POINTS

Pain	Cost	Current workaround
Driving out to gauge tanks and check wells	Remote service $55/mo (iWell, vendor); legacy SCADA $2,000-5,000/well (vendor)	Pumper visits; SCADA
Methane LDAR compliance on marginal wells	Industry claims ~300,000 wells could shut (advocacy) (CSU METEC)	Contractor surveys

Pain	Cost	Current workaround
Quarry dust exposure monitoring	Not found	Periodic sampling
Tank overfill and spills	Not found; operators climb tanks to gauge (Emerson, vendor)	Manual gauging

EXISTING PLAYERS

Company	Product	Price	Gap
iWell	Remote well and tank service	From $55/mo (vendor)	Subscription
Baker Hughes SentryNet, Emerson Rosemount, Digi/Innotek OverWatch, eLynx	Tank and well monitoring	Not found	Enterprise sales cycles; lock-in (inference)

IDEAS

1 TankEye Lite

16/25 [stated 17]
P/W/O/B/F 4/4/3/3/2

solar LoRa radar tank level with high-level alarm

B2B
Low-volume well sites; $300-500/unit + optional $10/mo

B2C
Rural owners of fuel/water tanks, $149

`Moderate-hard in hazardous areas; easy for non-hazardous B2C build · Radar level sensor, LoRa, solar`

2 Dust Tag

16/25 [stated 15]
P/W/O/B/F 3/3/3/4/3

optical particulate node for quarries (not a certified silica method)

B2B
Quarries, $200

B2C
Workshops, $79

`Moderate build · Optical PM sensor, LoRa`

3 Pumper Pager

16/25 [stated 14]
P/W/O/B/F 3/3/3/4/3

rugged vibration well-status node with local LED

B2B
Operators; price not specified

B2C
Not specified

`Moderate build · Accelerometer, LoRa, LED`

4 Acoustic leak finder

15/25 [stated 13]
P/W/O/B/F 2/3/3/3/4

handheld ultrasonic leak detector

B2B

Contractors, ~$300

B2C

Not specified

`Moderate build · Ultrasonic MEMS mics, DSP`

5 Methane Sniff Beacon

14/25 [stated 15]
P/W/O/B/F 4/3/3/2/2

fixed low-cost methane sensor on LoRa for LDAR screening

B2B

LDAR contractors and operators; price not specified

B2C

Not specified

`Hard (gas calibration, area ratings) build · Methane sensor, LoRa`

NAICS 22

Utilities

SUMMARY

Utilities are heavily regulated and buy slowly. Small co-ops, municipal utilities and tens of thousands of small water systems are underserved and short of budget. On the consumer side the opportunity is household resilience, such as outage awareness, rather than selling to utilities.

KEY FACTS

33,012 establishments, 856,308 employees, $507.7B value added (QCEW 22). US customers averaged **~11 hours of outages in 2024**, nearly double the prior-decade average (EIA). **42,199 small water systems** make up 82% of community water systems (2016, EPA OIG). Key rules: Lead and Copper Rule Revisions, with service-line inventories due Oct 16, 2024 (Illinois EPA). NERC CIP and UL/NEC for mains devices: no source retrieved; researchers advise avoiding grid-tied and life-safety products.

BUYERS

Segment	Count	How they buy
Rural co-ops and small municipal utilities	812 co-ops + 1,958 public power (2017, EIA)	General manager; distributors, state associations, conferences
Small water systems	~42,000 (2016)	Operator or board; grant-funded; circuit riders, rural water associations

Segment	Count	How they buy
Rural and off-grid households	Not found	Amazon or direct (B2C)

PAIN POINTS

Pain	Cost	Current workaround
Outages with no local awareness	~11 h/yr; cost per hour not found	Utility outage maps; generators
Small-system water loss, thin staffing	~1 in 5 gallons lost, $6.4B/yr (unverified, startup blog citing Bluefield)	Manual rounds; SCADA described as "million-dollar-plus" (WWD)
Lead service line inventory burden	Not found	Paper or software inventories
Voltage events damaging farm loads	Not found	Phase monitors

EXISTING PLAYERS

Company	Product	Price	Gap
Mission Communications	Cloud SCADA for small utilities	Not found	Price and complexity for small systems (vendor case studies)
High Tide Technologies	Utility monitoring	Not found	Same
120Water	Lead inventory software	Not found	Software only
Consumer outage/power monitors	Various	Not researched	Not researched

IDEAS

1 Pole Notice

18/25 [stated 17]
P/W/O/B/F 4/3/3/4/4

local-first outage and voltage-sag logger that chirps or texts

B2B
Co-ops buy in bulk for member programs, ~$50/unit

B2C
Home or farm, $79-129

`Moderate (isolated mains sensing via plug-in/clamp) build · ESP32, isolated voltage sensing, e-paper`

2 Leak Ear

17/25 [stated 15]
P/W/O/B/F 3/3/3/4/4

acoustic hydrophone logger for water mains

B2B

Small water systems, $300

B2C

Not specified

`Moderate build · Hydrophone, ADC, LoRa`

3 Tiny SCADA for Tiny Water

16/25
P/W/O/B/F 4/3/3/3/3

LoRa tank/pump/pressure node kit for systems under 500 connections

B2B

$400/site + optional $10/mo

B2C

Not specified

`Moderate build · LoRa nodes, pressure and level sensors`

4 Whole-farm meter display

16/25 [stated 14]
P/W/O/B/F 2/3/3/4/4

CT-clamp usage display, local only

B2B

Not specified

B2C

$129

`Easy build · CT clamps, ESP32, display`

5 LSL Field Tagger

14/25 [stated 11]
P/W/O/B/F 3/2/3/4/2

NFC handheld logger for lead-line inventory crews

B2B

Water systems; price not specified

B2C

None

`Easy (tiny market) build · NFC, handheld logger`

NAICS 23

Construction

SUMMARY

Money flows from owners to general contractors to specialty subcontractors through progress billing. Small trade contractors buy tools through distributors, big-box stores and online. The documented pains are tool and equipment theft, labor shortage and silica-dust compliance. Tool tracking is crowded with Milwaukee and Hilti, but cheap, brand-neutral, privacy-first jobsite sensing looks open.

KEY FACTS

944,595 establishments, 8,317,572 employees, $1,340.0B value added (QCEW 23). **92% of firms report difficulty hiring** and 45% report delays from worker shortages (NCCER/AGC 2025). Key rules: the OSHA silica rule sets a PEL of 50 µg/m³, and employers either follow Table 1 controls or measure exposure (Norton Rose Fulbright); OSHA noise limit of 85 dBA as an 8-hour TWA (OSHA).

BUYERS

Segment	Count	How they buy
Small trade contractors (electrical, HVAC, plumbing)	Not found	Owner decides; distributors and online
GCs and mid-size subs with equipment yards	Not found	Equipment or operations manager
Tool-heavy DIY and prosumers	Not found	Amazon (B2C)

PAIN POINTS

Pain	Cost	Current workaround
Labor shortage, poor jobsite visibility	Indirect; 45% report project delays	Overtime; subcontracting
Tool and equipment theft	~$1B/yr (old c.2007 NER estimate, weakly sourced); avg equipment theft $30,000 (unverified, Deep Sentinel, vendor); 55% had tools stolen in the prior year (DeWalt survey, vendor)	Gang boxes, ONE-KEY tags, camera services
Silica-dust compliance	OSHA estimate $659M/yr industry-wide; industry claimed up to $5B (ABC)	Table 1 controls, paper plans
Noise and heat exposure records	Not found	Paper records
Battery, charger and tool inventory chaos	Not found	Manual

EXISTING PLAYERS

Company	Product	Price	Gap
Milwaukee	ONE-KEY tags (Bluetooth only, phone-dependent)	10-pack $189.99 at one retailer (AirPinpoint, competitor blog)	No GPS; brand-locked
Hilti	ON!Track asset management	Quote-based (Capterra)	SaaS bundle
DeWalt	Tool Connect	Not found	Not found
Deep Sentinel-style services	Camera monitoring	Not found	Camera-based

IDEAS

1 Silica Table 1 logger

18/25
P/W/O/B/F 4/3/4/4/3

logs whether the shroud vacuum ran during cutting (a log, not an OSHA-grade measurement)

B2B
Contractors needing compliance records; $149 + report subscription

B2C
Hobbyist, $79

`Moderate build · Optical PM sensor, airflow sensor, ESP32`

2 Open jobsite theft tag

17/25
P/W/O/B/F 3/3/3/4/4

LoRa/BLE tags with a solar gateway in the gang box, no cloud

B2B
$15/tag, no subscription or optional $5/mo alerts

B2C
$39 kit

`Moderate build · ESP32/Nordic, LoRa, MEMS motion sensor`

3 mmWave trailer/gang-box intrusion alarm

17/25
P/W/O/B/F 3/3/2/5/4

no camera

B2B
$99/unit

B2C
Garages and vans, $79

`Easy-moderate build · 24/60 GHz radar, LoRa`

4 Noise/heat exposure badge

16/25
P/W/O/B/F 3/3/3/4/3

industrial-hygiene documentation, not medical

B2B
Price not specified

B2C
Not specified

`Moderate build · Mic, temp sensor, BLE`

5 Tool-crib shelf / battery board

15/25
P/W/O/B/F 3/3/3/3/3

weight or RFID charge-state board

B2B
Price not specified

B2C
Not specified

`Moderate build · Load cells or RFID, ESP32`

NAICS 31-33

Manufacturing (small and mid-size shops)

SUMMARY

Job shops sell parts to OEMs quote by quote. The owner or plant manager buys through industrial distributors and direct. Small shops run old machines at low utilization. Incumbent monitoring vendors are quote-priced SaaS aimed at bigger plants, which leaves room for a cheap retrofit sensor that keeps data local.

KEY FACTS

397,330 establishments, 12,784,104 employees, $2,930.1B value added (QCEW 31-33). **74% of manufacturers have fewer than 20 employees** (NAM). Job-shop spindle utilization is 30-45% (vendor, JITbase). Key rules: OSHA 85 dBA noise program; lockout/tagout and NFPA 70E (not sourced).

BUYERS

Segment	Count	How they buy
CNC job shops, 5-50 staff	Not found	Owner or operations manager; direct and online
Fabrication, molding and assembly plants	Not found	Maintenance manager
Garage and hobby CNC/laser users	Not found	Online (B2C)

PAIN POINTS

Pain	Cost	Current workaround
No visibility into spindle-on time on older machines	Utilization 30-45% (vendor)	Walk the floor; whiteboards
Unplanned downtime	~$1,840 per 3-hour stop (vendor example, unverified, Oxmaint)	Reactive repair
Labor shortage and cost	Avg wage $35.19/hr (IndustryWeek)	Overtime
Input cost and supply volatility	83.1% cite raw materials (NAM Q2 2026)	Not addressable by hardware
Noise, air quality, compressed-air waste	Not found	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
Guidewheel	Clamp-on power sensors + SaaS	Formerly $15,000/yr for 10 machines; ~$32K/plant/yr (competitor-authored, unverified)	Cloud, per-plant pricing

Company	Product	Price	Gap
		(PricingSaaS)	
MachineMetrics	Controller-data monitoring	Quote; ~$150-450/machine/mo (rival's claim)	Quote-only; aimed at larger shops
Scytec DataXchange	Monitoring software	$45/mo per machine (unverified)	SaaS
FourJaw (UK)	Retrofit monitoring	£36-90/machine/mo (FourJaw)	SaaS
Amper (now ECI)	Power-draw monitoring	Not found	Bundled into ERP

IDEAS

1 Clamp-on machine monitor

20/25
P/W/O/B/F 4/4/3/5/4

CT clamp with a shop-local dashboard

B2B
Job shops; $99/machine one-time + optional $5/mo

B2C
Hobby CNC/laser, $59

`Easy-moderate build · CT clamp, ESP32`

2 Andon/job beacon

18/25
P/W/O/B/F 3/3/3/5/4

LoRa stack light with e-ink job badges

B2B
Shops; price not specified

B2C
Not specified

`Easy build · ESP32, LEDs, e-ink, LoRa`

3 Cycle-counter puck

18/25
P/W/O/B/F 3/3/4/5/3

Hall/vibration part counter

B2B
Shops; price not specified

B2C
Not specified

`Easy build · Hall sensor, accelerometer`

4 Ultrasonic air/coolant leak detector

16/25
P/W/O/B/F 3/3/2/3/5

B2B
$349

B2C
$149

`Hard-ish (DSP) build · MEMS ultrasonic mics, DSP`

5 Shop noise logger badge

16/25
P/W/O/B/F 3/3/3/4/3

B2B
Price not specified

B2C
Not specified

Moderate build · Mic, BLE

NAICS 42

Wholesale Trade

SUMMARY

Wholesalers buy from manufacturers and resell, so their money sits in inventory. Inventory accuracy and forklift safety are real pains, but there is little sourced evidence and strong incumbents (WMS vendors, Zebra, RFID). It is the weakest of the B2B sectors for a solo hardware founder and is treated as under-researched.

KEY FACTS

637,241 establishments, 6,107,071 employees, $1,952.4B value added (QCEW 42). The 2025 value added is up 11% on 2024, as published by the mirror and not checked against bea.gov. Merchant wholesale inventories were $901.1B (Feb 2024) (TED Magazine). Key rules: OSHA 1910.178 for powered industrial trucks, the 7th most-cited standard in FY2018 (OSHA).

BUYERS

Segment	Count	How they buy
Independent distributors	Not found	Owner or operations
3PL-style warehouse operations	Not found	Warehouse manager
Small online resellers, home inventory	Not found	Online (B2C)

PAIN POINTS

Pain	Cost	Current workaround
Inventory accuracy and mispicks	Not found	Scanners, WMS
Forklift/pedestrian incidents	~85 deaths, 34,900 serious injuries/yr (older, unverified, SafetyNow)	Training, mirrors

Pain	Cost	Current workaround
Cold-chain and condition monitoring	Not researched	Not found
Labor cost	Not found	Not found
Tariff-driven cost volatility	Not found	Not addressable

EXISTING PLAYERS

Company	Product	Price	Gap
Zebra, Honeywell, Samsara	Scanners, RFID, telematics	Not found (names from researchers' background knowledge)	Not found

IDEAS

1 Weight-sensing shelf

18/25
P/W/O/B/F 3/3/3/5/4

load-cell bins for parts and hardware stock

B2B
Distributors and parts stores; price not specified

B2C
Maker version, $49

`Easy build · Load cell, HX711, ESP32`

2 Pick-to-light shelf strips

17/25
P/W/O/B/F 3/3/3/4/4

local-network LED bin strips

B2B
~$12/bin

B2C
Not specified

`Moderate build · ESP32, LED strips`

3 Open-protocol e-ink shelf labels

16/25
P/W/O/B/F 2/3/3/4/4

B2B
Price not specified

B2C
Not specified

`Moderate build · e-ink, BLE/LoRa`

4 Trailer/dock door and temperature logger

15/25
P/W/O/B/F 2/3/2/5/3

B2B
Price not specified

B2C
Not specified

`Easy (many rivals) build · Reed switch, temp sensor, LoRa`

5 Forklift pedestrian-proximity beacon

13/25
P/W/O/B/F 3/3/2/3/2

awareness aid, not a safety device

B2B
Price not specified

B2C
None

Hard (safety-adjacent) build · UWB/BLE, horn, light

NAICS 44-45

Retail Trade

SUMMARY

Retailers buy wholesale and sell to consumers on thin margins, so shrink comes straight off profit. Independent owners buy security and IT through POS resellers, Amazon and local installers. Theft is well documented but crowded with cameras and EAS. The open space is cold-chain alarms for small grocery and convenience stores, plus camera-free sensing.

KEY FACTS

1,071,548 establishments, 15,529,520 employees, $1,810.4B value added (QCEW 44-45). NRF 2026 reports shoplifting incidents **down 12.4%** in 2025, but organized retail crime is up at 40% of retailers. The sample is large chains (NRF). Key rules: FDA Food Code 41°F cold holding drives temperature logs; PCI-DSS and state biometric laws are reasons to avoid payments, audio and cameras (researchers' background, not sourced).

BUYERS

Segment	Count	How they buy
Independent boutiques and specialty stores	Not found	Owner; Amazon/Shopify ecosystem
Small convenience, grocery and liquor chains	Not found	Owner or ops manager; POS reseller, refrigeration contractor
Multi-site small chains (pharmacy, phone, vape)	Not found	Loss-prevention lead; security installers

PAIN POINTS

Pain	Cost	Current workaround
Theft and organized-crime shrink police don't pursue	68% of SMB retailers above 1.5% shrink (vendor-run survey, year unconfirmed, Security Magazine)	Cameras (54% added), locked cases
Walk-in cooler failure spoilage	$18k-$47k per event (vendor, GlacierGrid, Envigilance)	Manual logs, cheap Bluetooth thermometers
Gift-card and phone scams	42%/69% of retailers report increases (NRF)	Staff training; weakly addressable by hardware
Package theft	$12.8B-$15B/yr, estimates differ ~2x (Omnisend; SafeWise)	Doorbell cameras
Missing foot-traffic data	Not found as a documented pain	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
SensorPush	Sensor + G1 gateway	~$50 / ~$100 (2018, outdated)	Bluetooth range in steel walk-ins (inference)
Govee	Wi-Fi/BT thermometers	~$9-15	Connection and alert failures reported (Smart Home Explorer)
Kitchen Brains	ModulARM 75LC local walk-in alarm	Not found (Restaurant Supply)	Not connected
Temp Stick	Wi-Fi sensor, no subscription	~$139 (unverified)	Offline and text-alert anecdotes
Monnit, GlacierGrid, Envigilance	Sensors and monitoring services	Not found	Subscription service model
Verkada, Axon, Ring, Arlo	Cameras	Not found	Crowded; camera-based

IDEAS

1 Walk-in Watch

2/25 [adj from 20]
P/W/O/B/F 4/4/

LoRa/Wi-Fi door and probe node with local buzzer, optional text

B2B
Grocery, c-store, restaurant owners; $149 + $5-10/mo alerts or one-time

B2C
Chest-freezer owners, $59-79

Easy build · ESP32-C6 or nRF, NTC/DS18B20 probe, reed switch, piezo, AA cells

2 Quiet Counter

17/25
P/W/O/B/F 3/3/3/5/3

mmWave door/aisle counter, no camera

B2B
$79-129 + $5/mo dashboard

B2C
Home occupancy, $49

`Easy-moderate build · LD2410/C4001 mmWave, ESP32`

3 Back-Door Chime Pro

16/25
P/W/O/B/F 3/3/3/5/2

staff-door-ajar and entry alert

B2B
C-stores, $59

B2C
Door-ajar alert, $29

`Easy build · Reed/IR sensor, relay`

4 Porch Plate

15/25
P/W/O/B/F 3/2/3/4/3

weight/mmWave mat that alerts when a package lands or leaves

B2B
Small e-commerce sellers (weak)

B2C
$49-69

`Moderate (weatherproofing) build · Pressure sensor or mmWave, hub`

5 Shelf Count Puck

13/25
P/W/O/B/F 2/2/3/3/3

stock alert for high-theft items

B2B
Price not specified

B2C
Not specified

`Moderate-hard build · BLE, load cell`

[adj] Walk-in Watch: open space lowered from 3 to 2. Validation found local walk-in alarms (ModulARM) and subscription-free Temp Stick already on sale, so the "local buzzer, no subscription" gap is partly filled.

NAICS 48-49

Transportation and Warehousing

SUMMARY

Carriers earn per mile or per load on thin margins. Self-storage earns monthly rent per unit, and owners buy locks and gates through distributors. Cargo-theft losses jumped in 2025, but the ELD mandate makes trucking a certified-device category the founder should avoid. Self-

storage is the most attractive sub-segment: it is fragmented, driven by security, and dominated by a closed, quote-only incumbent (Janus/Nokē).

KEY FACTS

353,975 establishments, 7,532,742 employees, $1,008.9B value added (QCEW 48-49). 2025 cargo-theft losses were **~$725M (+60%)**, averaging $273,990 per theft (Verisk CargoNet). There are **50,000-58,000 self-storage facilities** (Modern Storage Media almanac). Key rules: ELDs under 49 CFR 395 have no small-carrier exemption (Overdrive); cellular devices need carrier certification, which pre-certified modules simplify (not sourced).

BUYERS

Segment	Count	How they buy
Independent self-storage owners	50,000-58,000 facilities; five public REITs hold 37.6% of space (SpareFoot)	Owner or manager; distributors (Janus/DoorKing resellers), trade shows
Owner-operators and small fleets	Not found	Owner; online, trucker forums, telematics reps
Couriers and small warehouses	Not found	Ops manager; Amazon, Grainger

PAIN POINTS

Pain	Cost	Current workaround
Cargo and trailer theft	$725M in 2025; $274k average	Hidden GPS ($14.99/mo), king-pin locks
Self-storage break-ins, shared gate codes	57% of operators deal with repeat break-ins (survey year unstated, Inside Self Storage); cost per incident not found	Smart locks, cameras, keypads
No alert when a unit door opens off-hours	Not found	Manual lock checks
Small-fleet telematics subscription cost	$25/vehicle/mo + $30 ELD (Linxup)	Cheaper trackers, phone apps
Dock temperature and door events	Not found	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
Janus / Nokē	Smart locks, Smart Entry	Quote-only; one ~~1,100-unit project ~$310K (~~$280/unit, may include more) (Janus, vendor)	Closed, costly

Company	Product	Price	Gap
PTI Security, OpenTech INSOMNIAC	Wired unit alarms; gateway for existing alarms	Not found	Wired retrofit cost
Generic door alarms	Per-door alarms	$50-250/door (StorageCafe estimate)	Batteries, false alarms
Linxup	Trackers, ELD	$25/vehicle/mo; asset tracker $14.99/mo	Subscription lock-in
Samsara, Motive	Telematics, ELD	Quote	ELD-centric

IDEAS

1 Unit Door Sentinel

20/25
P/W/O/B/F 4/4/3/5/4

magnet and accelerometer on each unit door reports opens and prying over LoRa to a site hub; no camera

B2B
Independent storage owners; $15-25/unit + $1/unit/mo, or per site

B2C
Tenants, $39-59 stand-alone alarm

`Moderate (200+ nodes, OTA) build · nRF/ESP32, SX126x LoRa, accelerometer, reed switch`

2 Gate Code Auditor

17/25
P/W/O/B/F 3/3/4/4/3

passive Wiegand logger flagging code sharing and tailgating

B2B
$199/site + $10/mo

B2C
HOA/private gates, $79

`Moderate (many controllers) build · Wiegand interface, ESP32`

3 Trailer Whisper

16/25
P/W/O/B/F 4/3/2/4/3

hidden LTE-M trailer tracker with BLE door sensor

B2B
Small fleets, $79 + $5-10/mo

B2C
RV/trailer owners, $59 + SIM

`Moderate build · nRF91/Quectel LTE-M, BLE`

4 Dock Door Pulse

15/25 [stated 13]
P/W/O/B/F 2/3/3/4/3

door-open and temperature logger

B2B
Small warehouses, vans; price not specified

B2C
Not specified

Easy build · Reed switch, temp sensor

5 Courier Drop Beacon

14/25 [stated 12]
P/W/O/B/F 2/2/3/4/3

BLE porch tag confirming drop time

B2B
Price not specified

B2C
Not specified

Easy (WTP unclear) build · BLE tag

NAICS 51

Information

SUMMARY

This is a large sector, but the slice that fits a solo hardware founder is small studios, podcasters and small server closets. Cheap alerting and studio status gear are the gaps.

KEY FACTS

300,282 establishments, 3,044,132 employees, $1,766.4B value added (QCEW 51). Comfort cooling is not designed for constant IT loads, and closets often lack backup cooling (Vertiv, vendor). Key rules: Section 889 bars federal grant recipients from covered telecom gear (eCFR 200.216). The FCC Covered List blocks new authorizations for foreign-made consumer routers (Mar 2026) and drones (Dec 2025); sources differ on exact dates (Cooley). Broadcast rules: not found.

BUYERS

Segment	Count	How they buy
Podcast and video studios, creators	Not found	Online, direct
Small broadcasters, church and community radio	Not found	Not found
SMB server rooms (often MSP-managed)	Not found	Through MSPs

PAIN POINTS

Pain	Cost	Current workaround
Unmonitored closet heat and water	US small-office downtime cost not found (only a UK £800-1,200/hr figure, unverified)	Cheap thermometers that don't alert on weekends (Level1Techs)

Pain	Cost	Current workaround
No automatic "recording live" signal	Not found	Manual switch by the door; users ask for OBS-linked lights (OBS forum)
Studio room status signage	Not found	Paper signs
Small-broadcast tally and cue gear	Not researched	Not researched

EXISTING PLAYERS

Company	Product	Price	Gap
APC	NetBotz temperature sensors	~$102 wired / ~$118 wireless (undated, CDW)	Needs NetBotz appliance; IT-shop pricing
AVTECH	Room Alert	Room Alert 12S $779 (Walmart); account $249.95/yr (2021)	Costly; SNMP/software
Vertiv	Geist Watchdog 15	Not found	Not found
Temp Stick	Wi-Fi sensor	~$149 (unverified)	Cloud-dependent
Orangemonkie, CBT Systems	On-air lights	Not found	Manual; not tied to record state

IDEAS

1 Closet Sentinel

2/25 [adj from 19]
P/W/O/B/F 3/4/

temp/humidity/leak puck with local web UI, buzzer, battery backup, alerts via webhook or ntfy

B2B
MSPs and SMBs; $79 + optional $5/mo relay

B2C
Homelab, ~$59

Easy build · ESP32, temp/humidity sensor, leak probe, buzzer

2 OnAir Puck

18/25
P/W/O/B/F 2/3/3/5/5

tally light that follows OBS, mic or DAW record state

B2B
Studio 3-pack

B2C
$59-89

Easy build · ESP32, USB, LEDs

3 Studio door sign set

17/25
P/W/O/B/F 2/3/3/5/4

e-ink/LED room status, local control

B2B
Price not specified

B2C
Not specified

Easy build · e-ink, ESP32

4 Rack-door environment strip

16/25
P/W/O/B/F 3/3/3/4/3

multi-point temp and airflow delta for 10-20U racks

B2B
Price not specified

B2C
Not specified

Moderate build · Temp sensors, airflow sensor

5 Podcast room noise-floor lamp

16/25
P/W/O/B/F 2/2/4/4/4

shows HVAC rumble and noise level

B2B
Price not specified

B2C
Not specified

Moderate build · Mic, ESP32, LED

[adj] Closet Sentinel: open space lowered from 3 to 2. Validation found the $145-$500 band already crowded and found no evidence that buyers want local-first.

NAICS 52

Finance and Insurance

SUMMARY

Insurers already pay for leak and freeze prevention through free kits and premium discounts, which makes them a distribution channel more than an end user. Bank-branch security is regulated and incumbent-heavy. The open lane is the home-device side that insurers subsidize, and validation suggests even that is commoditized.

KEY FACTS

582,814 establishments, 6,327,482 employees, $2,687.6B value added (QCEW 52). Water and freeze damage make up **~24-28% of homeowners claims at ~$14,000-15,400 each** (secondary sources; the primary III page was not retrieved) (Policygenius). Key rules: insurer discounts are state-filed rating factors, and bank branch security falls under the Bank Protection Act (neither researched).

BUYERS

Segment	Count	How they buy
Homeowners insurers and MGAs (Hippo, Chubb, Travelers, USAA)	Not found	Loss-prevention or innovation teams; bulk kits or retail subsidies
Commercial/multifamily brokers and risk engineers	Not found	Recommend devices to insureds
Small banks and credit union branches	~21,965 CU branches (June 2025, CU Times)	Integrators, core-vendor channels; low fit

PAIN POINTS

Pain	Cost	Current workaround
Water and freeze claims	~$14-15K average; ~1 in 60 insured homes/yr (secondary)	Free kits, discounts
Getting devices adopted and kept active	Hippo discounts avg $64/yr self-monitored, $91 pro-monitored (insurer self-reported)	Tie discount to device staying active
Frozen pipes in seasonal homes	Ice-dam claims ~$11,100 (Guidewire)	Thermostat alerts
Branch operations and security	Not researched	Not researched

EXISTING PLAYERS

Company	Product	Price	Gap
Hippo	Free smart-home kits (Kangaroo/SimpliSafe)	Free to policyholders	Data-sharing and cloud dependence (inference)
Chubb	WaterDefense; free Ting; StreamLabs discounts	5-7% discount for flow shutoff (2016 release, Chubb)	Homeowner-focused
Travelers	Wyze/SmartThings sensors on Amazon	Not found	Same
Consumer leak sensors (Govee, YoLink, Flo)	Leak sensors and shutoffs	Govee $13-18; YoLink $15-30; Flo shutoff $429-559 (affiliate sites, SmartHomeCompared)	Commodity

IDEAS

1 Proof-of-protection log

20/25
P/W/O/B/F 3/3/5/4/5

device keeps signed local logs the owner can show an insurer, no cloud account

B2B
Insurers accept attestations

B2C
$60-90

Moderate build · Secure element, ESP32/nRF, sensors

2 Insurer-grade freeze/leak node

3/25 [adj from 20]
P/W/O/B/F 4/

white-label sensor in the Vacancy Sentinel family

B2B
MGAs/insurers at wholesale

B2C
$40-80

Easy build · ESP32-C6/nRF, humidity/temp, probe pads

3 Seasonal-home freeze monitor

16/25
P/W/O/B/F 4/3/3/3/3

with cellular/LoRa gateway

B2B
Not specified

B2C
~$100 + optional service

Moderate (carrier cert) build · Pre-certified cellular module, temp

4 Pipe-clamp temperature probe

16/25
P/W/O/B/F 3/3/2/5/3

B2B
Not specified

B2C
Commodity pricing

Easy build · Clamp probe, BLE

5 Branch environment monitor

14/25
P/W/O/B/F 2/3/3/4/2

(closet, water)

B2B
Branches; price not specified

B2C
None

Moderate build · ESP32, sensors

[adj] Insurer-grade node: willingness to pay lowered from 4 to 3 and open space from 3 to 2. Validation found leak hardware at $13-$55 and insurers (Chubb, Hippo, Travelers) giving devices away, which depresses demand. The proof-of-protection log was not validated and keeps its score, but its demand is unproven.

NAICS 53

Real Estate and Rental and Leasing

SUMMARY

Money moves as rent and fees. Operating cost and vacancy are the profit levers, so owners buy devices that cut water loss, energy use and truck rolls. The researchers' strongest wedge was leak and freeze sensing in vacant units with a no-cloud angle for small landlords. Validation weakened it: hardware is cheap and landlord pain evidence was not found.

KEY FACTS

494,021 establishments, 2,441,949 employees, $4,234.1B value added (QCEW 53). Value added is inflated by imputed owner-occupied rent. NMHC cites **21.4M apartments in 5+ unit buildings** (older release) (NMHC). Key rules: landlord entry usually needs 24 hours' notice, varying by state (Nolo). Whether sensor data counts as "entry" is an open legal question, so designs should avoid audio, video and occupant identity.

BUYERS

Segment	Count	How they buy
Multifamily owners and operators	Not found	Regional ops and maintenance directors; proptech vendors
Small landlords and single-family rental owners	Not found	Amazon and direct; prefer one-time purchase
Property management companies	Not found	Per-door platforms; want PMS integration

PAIN POINTS

Pain	Cost	Current workaround
Water leaks in vacant and occupied units	One building saved $300,000 on renewal (vendor anecdote); Chubb up to 8% building discount (Habitat)	Weekly vacant-unit walk-throughs (BiggerPockets)
Vacant-unit HVAC and humidity waste	4-7x energy cost (vendor, unverified, Stratis)	Thermostat set points, staff checks
Turnover rekeying and lock batteries	Not found	Smart-lock codes; batteries every 4-6 months (vendor)
Truck rolls and unit-condition visibility	Not found	Periodic inspections

EXISTING PLAYERS

Company	Product	Price	Gap
Pro Sentry	Building leak detection	~$75/sensor + ~$1/device/mo; ~$350 per 1-bed	Sold to buildings and insurers, not small landlords
Dwelo, PointCentral, Stratis IoT	Vacant-unit smart-home platforms	Not found	Platform lock-in, cloud
Govee, YoLink, Flo, Phyn	Consumer leak sensors and shutoffs	$13-$700 (see NAICS 52)	Commodity; no landlord fleet management

IDEAS

1 Vacancy Sentinel

3/25 [adj from 20]
P/W/O/B/F 4/

battery leak, freeze and humidity puck, local-first

B2B
Property managers; per-door hardware + $1-2/mo dashboard

B2C
Small landlord or vacation home, $40-60

`Easy build · ESP32-C6/nRF, humidity/temp, probe pads, coin/AA cells`

2 Insurer-kit leak mesh

3/25 [adj from 19]
P/W/O/B/F 4/

local mesh with quiet e-ink hub matched to insurer requirements

B2B
Insurers/MGAs as discount kits

B2C
~$100 kit

`Moderate build · ESP32, LoRa, e-ink`

3 Offline BLE turnover lock-box

16/25
P/W/O/B/F 3/4/2/4/3

with logging

B2B
Price not specified

B2C
Not specified

`Moderate (crowded) build · nRF, lock mechanism`

4 Vacant-unit HVAC setback node

15/25
P/W/O/B/F 3/3/2/4/3

B2B
Per door

B2C
Modest

`Moderate build · Thermostat contact or Zigbee`

5 Equipment-rental asset tracker

14/25
P/W/O/B/F 2/3/2/4/3

(LoRa tags)

B2B

Price not specified

B2C

Not specified

`Moderate (pain not researched) build · LoRa tags`

[adj] Ideas 1 and 2: willingness to pay lowered from 4 to 3 and open space from 3 to 2, for the same reasons as NAICS 52. The only landlord forum post found was by a vendor. One snippet suggested landlords think a loud local alarm is enough (BiggerPockets).

NAICS 54

Professional, Scientific, and Technical Services

SUMMARY

Small law, accounting, architecture and engineering offices share the same hybrid-office problems: half-empty space, meeting rooms booked but unused, and unmonitored IT closets. Each firm is too small for enterprise occupancy tools.

KEY FACTS

1,731,519 establishments, 10,901,640 employees, $2,465.2B value added (QCEW 54).
Office occupancy is at **55.1% of the pre-2020 baseline**, with a 64.4% Tuesday peak (Kastle).
Key rules: workplace-monitoring and biometric privacy laws were not searched.

BUYERS

Segment	Count	How they buy
5-50 person professional firms	Not found	Office manager; online
Coworking and creative studios	Not found	Owner
MSPs reselling to those firms	Not found	Resale bundles

PAIN POINTS

Pain	Cost	Current workaround
Unused space and desks	Occupancy ~55%	Hot-desking
Ghost meeting-room bookings	24.5-28.1% of bookings (sensor study, vendor, Density); 17% at Columbia libraries (Columbia)	Teams panel check-in, which users report as flaky (Microsoft Tech Community)

Pain	Cost	Current workaround
IT closet heat	See NAICS 51	Cheap thermometers
Studio shoot-status and battery tracking	Not searched	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
Joan	E-ink room signs	Joan 6 ~$549; Pro ~$710-905; €49/mo + €9.99/device (older) (Capterra)	Per-device licence
Robin	Room booking	From $99/mo (comparison pages)	Subscription
Density	Waffle sensor	$149 + per-sq-ft subscription (2023, Business Wire)	Enterprise-oriented
Yealink	RoomSensor (PIR)	Not found	Pairs with own panels

IDEAS

1 Room puck

19/25
P/W/O/B/F 4/4/3/4/4

mmWave presence sensor with e-ink schedule that releases ghost bookings via Google/M365

B2B
Small offices; $120-180/room + $3/room/mo

B2C
Home-office variant, $99

`Moderate build · mmWave, e-ink, ESP32`

2 Closet Sentinel

2/25 [adj from 19]
P/W/O/B/F 3/4/

sold via MSPs (see NAICS 51)

B2B
$79 + optional $5/mo

B2C
~$59

`Easy build · As NAICS 51`

3 Desk-use counter

17/25
P/W/O/B/F 3/3/3/4/4

under-desk mmWave/PIR logger, CSV export, no cloud

B2B
$40/desk one-time

B2C
None

`Moderate build · mmWave/PIR, ESP32`

4 Privacy-first door headcount counter

17/25

P/W/O/B/F 3/3/3/4/4

with monthly email report

B2B

Price not specified

B2C

Not specified

`Moderate build · ToF/mmWave`

5 Shoot-live light + battery-rack monitor

17/25

P/W/O/B/F 2/2/4/5/4

for photographers

B2B

Price not specified

B2C

Not specified

`Easy build · ESP32, LEDs, current sensing`

The Teams thread supports demand for sensor-based release, but Microsoft said native support was "coming soon", so the gap may close. The score is unchanged.

NAICS 55

Management of Companies and Enterprises

SUMMARY

Corporate HQ and facilities buyers have the same hybrid-office problems as NAICS 54, with bigger budgets and much heavier security review. This was the least-researched sector. The researchers recommend selling through NAICS 54 channels and small facilities-management firms rather than targeting HQs directly.

KEY FACTS

100,043 establishments, 2,613,063 employees, $558.6B value added (QCEW 55). Class A+ buildings peaked at **78.8% occupancy** in Dec 2025 (Kastle). Key rules: corporate procurement typically requires SOC 2 questionnaires and bans unmanaged Wi-Fi devices (researchers' inference; not sourced).

BUYERS

Segment	Count	How they buy
Corporate real estate and workplace teams	Not found	Enterprise procurement
Facilities managers at multi-site firms	Not found	Approved vendor lists
IT/security teams (device veto)	Not found	Security review

PAIN POINTS

Pain	Cost	Current workaround
Space underuse	Occupancy ~55%	Consolidation
Ghost bookings	See NAICS 54	Panels
Multi-site closet monitoring	Not found (inferred)	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
Density, Robin, Yealink, Joan, APC	As NAICS 54 and 51	As listed there	Solo founder disadvantaged on security reviews

IDEAS

1 Ambient room-availability wall display

17/25
P/W/O/B/F 2/3/3/5/4

B2B
FM teams; price not specified

B2C
None

Easy build · e-ink/LED, ESP32

2 Room puck

16/25
P/W/O/B/F 3/3/2/4/4

(as NAICS 54)

B2B
Per room

B2C
None

Moderate build · As NAICS 54

3 Desk counter

16/25
P/W/O/B/F 3/3/2/4/4

(as NAICS 54)

B2B
Per desk

B2C
None

Moderate build · As NAICS 54

4 **Multi-site closet monitor** 2/25 [adj from 17]
P/W/O/B/F 3/4/

with simple dashboard

B2B
Price not specified

B2C
None

Moderate build · As NAICS 51

[adj] Multi-site closet monitor: open space lowered from 3 to 2 on the same crowded-market evidence as Closet Sentinel.

5 **Offline floor-use "heat" display** 15/25
P/W/O/B/F 2/2/3/4/4

B2B
Price not specified

B2C
None

Moderate build · mmWave, display

NAICS 56

Administrative and Support and Waste Management

SUMMARY

These are large, fragmented service businesses with field crews (guards, janitorial, landscaping, pest control, waste hauling) that sell labor-hours or recurring routes. Proof that service happened is a paid software category, but it relies on phones. The best hardware gaps are cheap verification tags and bait-station or dumpster sensors, where incumbents quote prices only on request.

KEY FACTS

694,633 establishments, 9,213,984 employees, $919.9B value added (QCEW 56). Structural pest control had **$13.4B revenue in 2025 across 16,565 firms**, 81.4% with one or two locations (NPMA). Key rules (researchers' background, not sourced): state guard licensing; FIFRA and state tamper-resistance rules for bait stations; state notice laws for GPS tracking of employees (CT, NY, DE).

BUYERS

Segment	Count	How they buy
Landscaping and lawn firms	~690,000-730,000 businesses (secondary, conflicting) (NALP)	Owner; trade shows, software resellers
Pest control firms	16,565	Owner; PestWorld, pest supply distributors
Security guard companies	~9,500 establishments (First Research, year unstated)	Owner; software vendors
Janitorial firms, waste haulers	Not found	Not found

PAIN POINTS

Pain	Cost	Current workaround
Proving service happened ("you didn't mow")	Lost payment per dispute; dollar figure not found	GPS trackers ($25/vehicle/mo), crew photos, route sheets (LawnSite)
Guard patrol verification and tag spoofing	Not found	NFC/QR tags + app; tags can be copied (Miratag)
Manual bait-station route checks	Not found	Manual inspection
Overflowing or underfilled dumpsters	Extra pickup **$149-$198** each (LA County; ECUA)	Compology/Enevo; route checks
Restroom consumables	Not found	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
PatrolTech	Guard tour app	$29/mo for 10 checkpoints + $2 each extra (Slashdot)	Phone-centric; copyable tags
Silvertrac	Guard management	~$249/mo for 10 devices (Capterra); setup fees $842-1,404 (competitor claim)	Cost, minimums
Guard1	Guard tour hardware	~$995-1,385 packages (aggregator)	Quote-based
Linxup, GPS Insight	Vehicle GPS	$25/vehicle/mo	No site-level proof
Jobber, Aspire, LMN	Field-service software	Not found	Phone check-ins

Company	Product	Price	Gap
Compology, Enevo	Dumpster fill sensors	Quote-only, per container per month	Small haulers priced out (inference)

IDEAS

1 Crew Beacon Tag

4/25 [adj components; total unchanged]

tamper-evident BLE + NFC fob per client site that logs a signed, non-clonable visit token

B2B

Landscape, janitorial, guard firms; $12-20/tag + $29/mo for 25 sites

B2C

Homeowners log cleaner/lawn visits, $25

Easy build · nRF52, NFC-A coil, secure element

2 Dumpster Level Puck

16/25 [stated 17]
P/W/O/B/F 3/4/3/3/3

ToF/ultrasonic fill sensor on LTE-M

B2B

Small haulers, property managers; $89 + $4/mo

B2C

HOA/farm bins

Moderate-hard build · nRF9160 or pre-cert LTE-M, ToF/ultrasonic

3 Bait Station Check Tag

16/25
P/W/O/B/F 3/3/4/3/3

reports lid opens and station status over BLE

B2B

Pest firms, $15-25/station

B2C

Gardeners, $29

Moderate (tamper-resistance rules) build · Reed/Hall sensor, BLE

4 Round Keeper guard hub

15/25
P/W/O/B/F 3/3/2/4/3

pocket key with haptic prompts, offline tour log

B2B

Price not specified

B2C

None

Moderate (competes with free apps) build · nRF, haptics

5 Restroom refill counter

13/25 [stated 11]
P/W/O/B/F 2/3/3/3/2

B2B

Price not specified

B2C

None

Moderate build · Level sensor, BLE

[adj] Crew Beacon Tag: pain raised from 3 to 4 because LawnSite has repeated threads on non-payment disputes. Willingness to pay lowered from 4 to 3 because operators already settle disputes with free phone photos and existing GPS. The total stays at 19.

NAICS 61

Educational Services

SUMMARY

Schools buy sensors for safety and facilities through district procurement, often with grants or settlement money. Vape detection is the proven category. False alarms, recurring licences and student privacy are the documented weaknesses of incumbents.

KEY FACTS

242,679 establishments, 13,195,807 employees, $353.2B value added (QCEW 61). QCEW includes public schools; the private-only CBP count is 118,004. There are **13,318 regular districts** and 91,390 traditional public schools (EdWeek). Key rules: the amended COPPA rule took effect June 23, 2025, with compliance due April 22, 2026, and biometrics now count as personal information (Clym, secondary); FERPA. Design implication: no audio and no student identifiers.

BUYERS

Segment	Count	How they buy
K-12 districts	13,318	Facilities, IT or safety director; board approval; pilots in a few schools
Colleges and universities	~3,931 Title IV degree-granting (uncertain reading, NCES)	Residence life, facilities
Makerspaces, trade schools, tutoring	Not found	Direct

PAIN POINTS

Pain	Cost	Current workaround
Vaping in restrooms	Median **$925/detector**, median contract $14,204 across 284 districts (vendor-analysis, CivicIQ)	HALO, Zeptive, Soter, Triton
False alarms and privacy	"Hundreds, sometimes thousands" of alerts a day in Montgomery County (FOX 5 DC); student searched after deodorant set off an alert (NPR)	Calibration; staff checks
Subscriptions and lock-in	Software in 204 of 284 awards; renewals ~$39-150/unit/yr	Multi-year renewals

Pain	Cost	Current workaround
Air quality and CO2	Not researched	Not researched

EXISTING PLAYERS

Company	Product	Price	Gap
Motorola HALO	Vape/aggression sensor	$964-$3,050/unit + licence	False alarms, subscription
Triton	Vape sensor	~$750-999 incl. 1 yr, then ~$110-150/yr (Triton, vendor)	Subscription
Zeptive	Vape sensor	~$1,100; in 195 awards	Not found
Soter	Vape sensor	>$12,000 + $1,500/yr per school (WKBW)	Cost
Vape Guardian	Vape sensor	Not found	Claims false-alert fix (press release)

IDEAS

1 Quiet-Room aerosol sensor

19/25
P/W/O/B/F 4/4/3/4/4

PM/VOC sensing, no mic, no cloud, local alert; flat price

B2B
Districts; $300-500/unit one-time, no mandatory licence

B2C
Parents and apartment owners, $80-120

`Moderate build · ESP32, Sensirion-class PM, VOC, optional mmWave`

2 Classroom CO2 and noise display

18/25
P/W/O/B/F 3/3/2/5/5

design-led desk object

B2B
Bulk to schools

B2C
$60-90

`Easy build · CO2 sensor, mic, display`

3 Makerspace tool-use and air-filter monitor

18/25
P/W/O/B/F 2/3/4/5/4

B2B
Price not specified

B2C
Not specified

`Easy-moderate build · CT clamp, PM sensor`

4 Restroom occupancy counter

17/25
P/W/O/B/F 3/3/3/4/4

(mmWave, no image)

B2B
Price not specified

B2C
None

Moderate build · mmWave

5 Bell/PA hardware

12/25
P/W/O/B/F 2/3/2/3/2

B2B
Price not specified

B2C
None

Moderate (crowded) build · Audio amp, network

Validation strengthens the pain and willingness-to-pay case for idea 1: settlement money is one-time, which suits a flat price. It also warns that false alarms are a sensing-quality problem, so a no-mic device must still out-discriminate HALO. No evidence was found that districts value no-cloud. The score is unchanged.

NAICS 62

Health Care and Social Assistance (non-medical slice only)

SUMMARY

Only childcare, social assistance and senior-living facility operations were screened; medical products are excluded. Childcare centers have very thin margins and buy cheap operational tools. Senior-living operators have compliance-driven water-temperature logging needs, which are larger budgets but close to regulation.

KEY FACTS

2,039,908 establishments, 24,311,493 employees, $2,327.2B value added for the whole sector, mostly medical (QCEW 62). Child day care (NAICS 6244) has **79,708 establishments** (QCEW 2024; CBP 2023: 82,162) (QCEW 6244). Child Care Aware counts 92,613 licensed centers and 98,807 family child-care homes (FFYF). Key rules: CMS requires water-management programs (ASHRAE 188) for certified facilities (Envigilance, vendor). Infant ratios are 3-6 per staff member (Niskanen). Anything resembling child supervision or life safety is excluded.

BUYERS

Segment	Count	How they buy
Child care centers	~79,700-92,600	Director or owner; Amazon, supply catalogs
Family child care homes	~98,800	Consumer-like
Senior living operators	Not found (~621,000 assisted living units occupied, Senior Housing News)	Maintenance directors; corporate vendor lists

PAIN POINTS

Pain	Cost	Current workaround
Rising costs, thin margins	65% of centers raised tuition (FFYF); wages 56-68% of costs	Tuition increases
Staffing and ratio compliance	Not found	Manual headcounts
Senior-living water-temperature logging	Monitoring services "from $750/mo" (vendor)	Manual readings
Senior-living energy and HVAC	Not found	Not found
Classroom air quality	Not found	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
Envigilance	Water-monitoring service	From $750/mo (vendor)	Expensive for small operators
Watts	Recirculation control	Not found	Not found
JUMO	Wireless temperature loops	Not found	Not found
Brightwheel, Procare	Childcare software	Not found	Software, not hardware

IDEAS

1 Daycare room CO2/humidity/noise display

19/25
P/W/O/B/F 3/3/3/5/5

friendly design-led object

B2B
Centers, $80-120

B2C
Parents and home daycares, $60

Easy build · CO2, humidity, mic, display

2 Pipe-temperature logger tag

18/25
P/W/O/B/F 3/4/4/4/3

clamp-on probe logging loop temperatures over LoRa, with a printable log (non-safety)

B2B
Senior living, $60-120/node + optional $5-10/mo

B2C
Homeowners with recirculation, $50

`Moderate build · Clamp probe, LoRa`

3 Quiet-hours noise lamp

18/25
P/W/O/B/F 2/2/4/5/5

for classrooms

B2B
Price not specified

B2C
Yes

`Easy build · Mic, RGB LED`

4 Kitchen fridge/freezer logger

17/25
P/W/O/B/F 3/4/2/5/3

B2B
Price not specified

B2C
Not specified

`Easy (crowded) build · Temp probe`

5 Door-ajar chime

17/25
P/W/O/B/F 3/3/3/5/3

(convenience only, not life safety)

B2B
Price not specified

B2C
Not specified

`Easy build · Reed switch`

NAICS 71

Arts, Entertainment, and Recreation

SUMMARY

This is a membership- and ticket-driven sector. Fitness is the best-documented part and is growing; climbing gyms have flat revenue and fewer openings. Operators are small and short-staffed, and unstaffed 24/7 access is a big theme. Access control is crowded and cloud-based, so the gaps are niche, local-first hardware.

KEY FACTS

191,265 establishments, 2,952,445 employees, $321.3B value added (QCEW 71). **81M Americans held gym memberships in 2025** (Athletic Business). There are 574 US climbing gyms (IBISWorld estimate), and 73% of operators report a worsening economy (Athletic Business/CBJ). Key rules: ADA, state health-club contract laws and venue noise ordinances (not found).

BUYERS

Segment	Count	How they buy
Gym and studio owners	Not found	Through management-software ecosystems (Mindbody, Gymdesk) with access partners (Athletech)
Climbing gym operators	574	Owner; trade body CBJ
Venues, marinas, bowling, golf, museums	Not found	Not found

PAIN POINTS

Pain	Cost	Current workaround
Staffing limits on hours; unstaffed access tied to payment	Not found (vendor claims only)	Kisi/Brivo access control
Climbing gym economic pressure	73% report worsening conditions	Cost cutting
Lost keys and rekeying	$1,500 for 10 doors (vendor hypothetical)	Rekey
Unstaffed-hours security	Not found	Cameras

EXISTING PLAYERS

Company	Product	Price	Gap
Kisi, Brivo	Cloud access control	Not found	Cloud-based; crowded
Gymdesk, GymMaster, Rezerv/igloohome	Software + door integration	Not found	No same-day lockout after failed payment (Gymdesk)

IDEAS

1 Climbing comp timer/leaderboard display

20/25
P/W/O/B/F 3/3/4/5/5

B2B
Gyms, $199-399 one-time

B2C
Home-wall users, $79

Moderate build · ESP32, HUB75 LED or e-ink, BLE

2 Hangboard force logger

19/25
P/W/O/B/F 2/4/4/4/5

load-cell plate, local display, no app account

B2B
Gyms, $149

B2C
Climbers, $99

`Moderate build · HX711, load cell, nRF, e-ink`

3 Venue SPL logger/sign

19/25
P/W/O/B/F 3/3/4/5/4

sound-level display and log as ordinance proof

B2B
Venues, $199 + log export

B2C
Practice rooms and bands, $99

`Easy build · ESP32, calibrated MEMS mic`

4 Offline NFC member door reader

17/25
P/W/O/B/F 4/4/2/4/3

validates a signed list locally

B2B
Small gyms, $120 + $5/mo sync

B2C
None

`Moderate build · ESP32, PN532 NFC, relay`

5 Weight-room occupancy strip

17/25
P/W/O/B/F 3/3/3/4/4

mmWave count shown on lobby display

B2B
Gyms, $149

B2C
None

`Moderate build · LD2450 mmWave, ESP32`

NAICS 72

Accommodation and Food Services

SUMMARY

Restaurants run on thin margins and heavy compliance paperwork. Short-term rentals are a host-run market where Airbnb's 2024 indoor-camera ban created a legitimate niche for privacy-safe sensors. Restaurant operators buy through dealers and online; STR hosts buy on Amazon, direct and through PMS integrations.

KEY FACTS

793,370 establishments, 14,206,569 employees, $965.3B value added (QCEW 72). **42% of restaurant operators were unprofitable in 2025** (NRA). There are ~2.46M STR listings and 785K hosts (2023, secondary, Airbtics). Key rules: FDA Food Code requires cold food at 41°F or below and hot at 135°F or above (Xenia, secondary). Airbnb allows decibel-only noise monitors in common areas, disclosed in the listing, with no sound recording (Hostfully).

BUYERS

Segment	Count	How they buy
STR hosts and small property managers	~785K hosts (2023)	Amazon, direct, PMS integrations; Airbnb subsidizes Minut (Airbnb)
Independent restaurants	Not found (independents fell 2.3% in 2025, NRN)	Owner; dealers and online
Chains and multi-unit operators	263,000+ chain locations	Corporate procurement (ComplianceMate, Monnit)

PAIN POINTS

Pain	Cost	Current workaround
Walk-in refrigeration failure	$6,550-$40,000 per incident (vendor, GlacierGrid)	Manual 4-hour logs, paid sensors
Thin margins	42% unprofitable	Any product must pay back fast
STR parties and noise	~$1,200 per party (Minut-linked, marketing); one host went from 4-5 neighbor complaints a year to zero with alerts (STR Specialist)	Noise sensor + subscription
Temperature-log paperwork	Not found	Paper logs
Subscription cost of monitors	Minut price creep and billing disputes (Trustpilot)	Tolerate or cancel

EXISTING PLAYERS

Company	Product	Price	Gap
Minut (owns NoiseAware, unverified)	Noise/occupancy sensor	$150 + $10/mo/device	Billing complaints; device useless without plan
NoiseAware	Noise sensor	$199 + $9/mo	Plug-in only, bulky
Alertify	Noise/smoke/vape sensor	$150 + $6/mo	Less polished app

Company	Product	Price	Gap
Party Squasher, Roomonitor	Device counting; monitoring	Not found	No sound sensing; Europe-focused
Monnit	Food-service sensors	From $49, free basic monitoring (Monnit)	Cheap competitor
ComplianceMate	Food-safety sensors	$650 for 2 sensors incl. 1 yr (listing inconsistent)	Cost
SensoScientific	Temperature monitoring	Quote	Quote-only

IDEAS

1 Local-only STR noise + occupancy puck

2/25 [adj from 21]
P/W/O/B/F 4/4/

decibel-only, no audio stored, local hub, one-time price, Airbnb-compliant

B2B
Small PMs; $79-129/unit + optional $2/mo SMS relay

B2C
Hosts, $99-149 one-time

`Easy-moderate build · ESP32-S3, MEMS mic (on-chip dB), LD2410 mmWave, optional CO2/vape`

2 Bar/cafe sound-level sign

20/25
P/W/O/B/F 3/3/4/5/5

public dB display to stay under ordinance

B2B
Bars and cafes, $129

B2C
Home studio, $79

`Easy build · ESP32, I2S mic, LED matrix`

3 STR quiet-hours nudge lamp

20/25
P/W/O/B/F 3/3/4/5/5

glows amber or red over a dB threshold, no alerts

B2B
Hosts, $89

B2C
Apartment dwellers, $59

`Easy build · ESP32, mic, RGB LED`

4 Walk-in door-ajar + temperature tag

18/25
P/W/O/B/F 4/3/2/5/4

with local siren/display

B2B
Independent restaurants; hardware + $5/mo SMS

B2C
Chest freezer owners, $35

`Easy build · nRF52/ESP32, NTC probe, reed switch, e-ink`

5 Health-code temp-log display

17/25
P/W/O/B/F 3/3/3/4/4

e-ink logger that exports the daily 41°F/135°F log

B2B	B2C
$149 + free export	None

`Moderate build · nRF, probe, e-ink, USB`

[adj] STR puck: open space lowered from 3 to 2. Minut reportedly owns NoiseAware and Airbnb gives hosts a free Minut sensor, and Minut already markets no audio recording, so privacy alone does not differentiate. The one-time price and the absence of cloud lock-in do. The validation note found no evidence that buyers want headcount.

NAICS 81

Other Services (except Public Administration)

SUMMARY

These are fragmented, owner-operated businesses (auto repair, laundromats, salons, car washes, churches, pet care) paid per transaction or appointment. Laundromat and car-wash electronics are locked into OEM ecosystems. Retrofit add-ons that read machine status may be open.

KEY FACTS

886,613 establishments, 4,705,257 employees, $629.7B value added (QCEW 81). Auto repair shop counts conflict (~166K-284K) (Orbital). There are about **10,890 laundromat and drycleaner establishments** (CBP 2023 via Vanta, approximate). The pet "other services" segment is worth $14.3B (APPA). Key rules: PCI, state licensing and EPA refrigerant and wash-water rules (not found).

BUYERS

Segment	Count	How they buy
Laundromat owners	~10,900	Distributors tied to equipment makers (Automatic Industries)
Independent auto shops	~166K-284K (conflicting)	Parts and tool distributors, PartsTech-style platforms
Pet groomers/boarding, salons, churches	Not found	Not found

PAIN POINTS

Pain	Cost	Current workaround
Technician shortage in auto repair	31% call it their biggest challenge (S&P/PartsTech)	Not hardware-addressable
Laundromat utilities and equipment replacement	Cited by 55% and 53% (PlanetLaundry)	Card systems
Laundromat theft and vandalism	Not found (vendor claims)	Card readers, cameras
Car-wash equipment downtime	Not found	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
Coinamatic, Automatic Industries	Laundry card systems	Not found	OEM ecosystem
PartsTech, Fullbay	Shop software	Not found	Software only
Salon, pet and church hardware players	Not found	Not found	Not found

IDEAS

1 Clamp-on washer/dryer cycle sensor

19/25
P/W/O/B/F 3/4/3/5/4

retrofits any machine; status for owner and customers

B2B
Laundromats, $25-40/machine + $1/mo

B2C
"Laundry done" puck, $39

`Easy build · ESP32-C3/nRF, accelerometer, CT clamp`

2 Church/venue sound-level and HVAC nudge panel

18/25
P/W/O/B/F 2/3/4/5/4

B2B
$149

B2C
Home studios, $79

`Easy build · ESP32, mic, e-ink`

3 Kennel noise/temperature tag

18/25
P/W/O/B/F 3/3/3/5/4

local alerts, no cameras

B2B
Boarding facilities, $79/unit

B2C
Pet crate/car, $49

`Easy build · nRF/ESP32, temp sensor, mic`

4 Shop-bay air/tool-use monitor

17/25
P/W/O/B/F 3/3/4/4/3

tallies bay utilization

B2B
Auto shops, $99 + dashboard

B2C
Garage hobbyists, $49

`Moderate build · ESP32, CT, pressure sensor`

5 Salon/barber e-ink door sign

16/25
P/W/O/B/F 2/2/3/5/4

tap-to-update "back in 10"

B2B
$79

B2C
Home businesses, $59

`Easy build · ESP32, e-ink`

NAICS 92

Public Administration

SUMMARY

Local governments are numerous and spend documented money on visitor counters for parks and libraries. Procurement is slow and bid-based. Federal grant rules (Section 889) favor trusted suppliers with compliant supply chains. Current counters cost $150 to $6,000 per unit.

KEY FACTS

139,927 establishments, 7,846,751 employees; BEA value added $3,428.6B covers all government, not just NAICS 92 (QCEW 92). There were **91,438 local governments in 2025**, 57.7% of them school or special districts (Census). Key rules: 2 CFR 200.216 / Section 889 bans covered telecom and surveillance gear for grant recipients, with no waivers (Taft). Check every module vendor's status before bidding. Procurement thresholds, cooperative contracts and Buy American rules were not found.

BUYERS

Segment	Count	How they buy
Public libraries	Not found	Library director or IT; often via consortium (e.g. Minitex cooperative pricing, Minitex)
Parks, recreation and trails	Not found	Regional agencies share accounts; bids

Segment	Count	How they buy
Public works and small municipalities	Within 91,438 local governments	Council-approved bids above local thresholds

PAIN POINTS

Pain	Cost	Current workaround
Trail and park visitor counting	$4,000-6,000/unit + $35/mo data + $125 battery every 2 years (MORPC; Lake County bid)	Eco-Counter; shared units
Library door-count cost and accuracy	$150-800/unit; $550 called "steep" (LIBRARYDIR)	SenSource, Axis, Geovision
Public-works remote monitoring	Not found (inferred)	Not found
Meeting-room usage	Not found	Not found

EXISTING PLAYERS

Company	Product	Price	Gap
Eco-Counter	Urban Post MULTI	$5,450 (2021 quote) + $35/mo	Price
SenSource	Library door counter	$149.95 wired	Wired
Axis, Geovision	Camera-based counters	~$500-800 (2023 list-serv)	Camera-based
Chambers Electronics	Park counters	Not found	Not found

IDEAS

1 Trail/park counter

19/25
P/W/O/B/F 4/4/4/3/4

solar, local logging, PIR/mmWave, no mic, SD export, design-led

B2B
Parks and trail agencies, $400-700

B2C
Trail clubs, HOAs

`Moderate build · PIR/mmWave, solar, SD, optional LoRa`

2 Library door counter

18/25
P/W/O/B/F 3/4/3/5/3

battery, USB report

B2B
Libraries, $150-250

B2C
None

`Easy build · ToF/PIR, nRF, USB`

3 Meeting-room sign with auto-release

17/25
P/W/O/B/F 3/3/3/4/4

for city halls and libraries

B2B
Price not specified

B2C
None

`Moderate build · mmWave, e-ink`

4 Live capacity sign

17/25
P/W/O/B/F 2/3/4/4/4

for parks and pools, fed by counter

B2B
Price not specified

B2C
None

`Moderate build · Display, LoRa`

5 Public-works LoRa level/temperature sensor

16/25
P/W/O/B/F 3/3/4/3/3

(non-safety use)

B2B
Price not specified

B2C
None

`Moderate build · LoRa, level/temp sensors`

How to read this screen

Each sector section follows the same layout: Summary, Key facts, Buyers, Pain points, Existing players, Ideas. **Establishments, employment and value added come from one source for every sector**: BLS QCEW 2024 annual averages (all ownerships summed) and BEA 2025 value added via the DBnomics mirror (QCEW open data; BEA via DBnomics). QCEW excludes the self-employed and most farm operators, so sectors 11 and 92 are undercounted. BEA's real-estate value added includes imputed rent on owner-occupied housing, and its "Government" line covers all government, not just NAICS 92. Figures from these two sources supersede any sector size in the other notes.

Scores are the researchers' judgments on five 1-5 criteria: **P**ain, **W**illingness to pay, **O**pen space, **B**uildable solo, founder **F**it. In several sectors the researchers' stated totals did not equal the sum of their component scores (sectors 11, 21, 22, 48-49 and 56). This report uses the sum of the components and shows the stated total in brackets. Scores were changed only where the validation notes gave evidence, marked **[adj]** and explained. Labels used throughout: **(vendor)** means the figure comes from a company selling in the category; **(unverified)** means a secondary or snippet-level source the researchers could not confirm; **not found** means the research looked and came up empty. Reddit and most trade forums

were not accessible to any researcher, so every "pain" row rests on trade press, surveys, vendor material or local news unless stated otherwise.